



--	--	--	--	--	--	--	--	--	--

AB Quest

(ADHYANT BRAIN QUEST)

For

SPARK COURSE

Time : 2 Hrs.

(For IX to X Moving Students)

Maximum Marks : 320

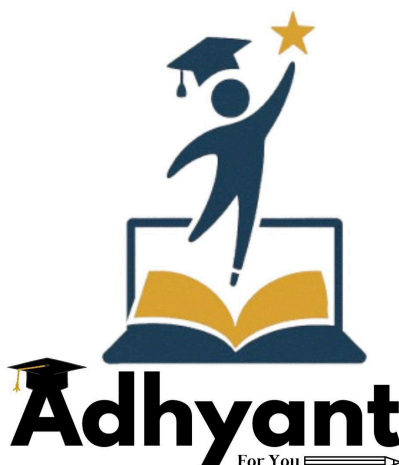
Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

Things NOT ALLOWED in EXAM HALL : Blank Paper, clipboard, log table, slide rule, calculator, camera, mobile and any electronic or electrical gadget. If you are carrying any of these then keep them at a place specified by invigilator at your own risk

INSTRUCTIONS

1. This booklet is your Question Paper. **DO NOT** break seal of Booklet until the invigilator instructs to do so.
2. Fill your Form No. in the space provided on the top of this page.
3. The Answer Sheet is provided to you separately which is a machine readable Optical Response Sheet (ORS). You have to mark your answers in the ORS by darkening bubble, as per your answer choice, by using black & blue ball point pen.
4. Total Questions to be Attempted 80. Part-I : 20 Questions & Part-II : 60 Questions.
5. After breaking the Question Paper seal, check the following :
 - a. There are **11 pages** in the booklet containing question no. 1 to 80 under 2 Parts i.e. Part-I & Part-II.
 - b. Part-I contains total 20 questions of IQ (Mental Ability).
 - c. Part-II contains total 60 questions under 4 sections which are - Section (A) : Physics, Section (B) : Chemistry, Section(C) : Biology & Section (D): Mathematics.
6. Marking Scheme:
 - a. If darkened bubble is RIGHT answer : **4 Marks**.
 - b. If no bubble is darkened in any question: **No Mark**.
 - c. **Only for part - II** : If darkened bubble is WRONG answer: **-1 Mark (Minus One Mark)**.
7. Think wisely before darkening bubble as there is negative marking for wrong answer.
8. If you are found involved in cheating or disturbing others then your ORS will be cancelled.
9. Do not put any stain on ORS and hand it over back properly to the invigilator.

NOTE :- Return this Test Paper



PART-I

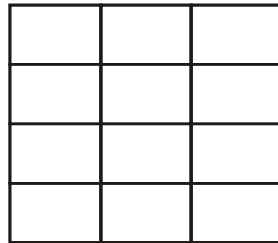
IQ (MENTAL ABILITY)

This section contains **20 Multiple Choice Questions**. Each question has four choices (1), (2), (3) and (4) out of which ONLY ONE is correct.

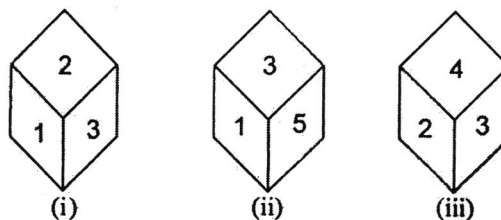
1. Rani and Shreshtha are a married couple having two daughters, Medha and Deepti. Deepti is married to Anurag who is the son of Garima and Tarun. Nidhi is the daughter of Anurag. Komal, who is Anurag's sister, is married to Harshit and has two sons, Aman and Prem. Prem is the grandson of Garima and Tarun.

What is the relationship between Aman and Nidhi ?

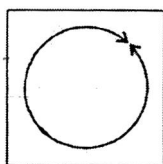
- (1) Cousins (2) Husband-wife (3) Father-daughter (4) Uncle-niece
2. If the word 'MECHANIC' is coded as 'EMHCNACI' in a certain code language, then how will 'WESTSIDE' be coded in the same language ?
- (1) EWTSISED (2) EWTSSIED (3) EWTSISDE (4) WETSSIED
3. Find the number of squares formed in figure ?



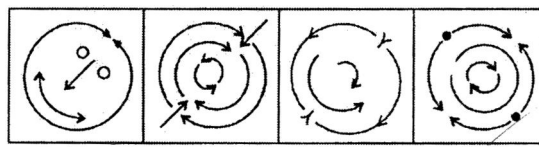
- (1) 22 (2) 18 (3) 20 (4) 30
4. A dice is thrown three times and its three different positions are given below. Find the number on the face opposite the face showing 3.



- (1) 1 (2) 4 (3) 5 (4) 6
5. Rekha moved a distance of 75m towards the North. She then turned to the left and walked for about 25m turned left again and walked 80m. Finally she turned to the right at an angle of 45° . In which direction was she moving finally ?
- (1) North-east (2) North-west (3) South (4) South-west
6. In the following question, you are given a figure (X) followed by four alternative figures (1), (2), (3) and (4) such that fig. (X) is embedded in one of them. Trace out the alternative figure which contains fig. (X) as its part.



(X)



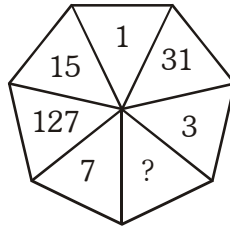
(1)

(2)

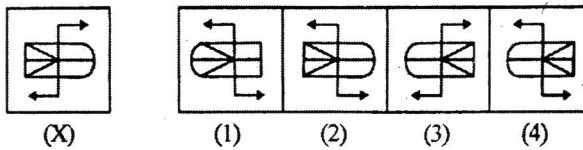
(3)

(4)

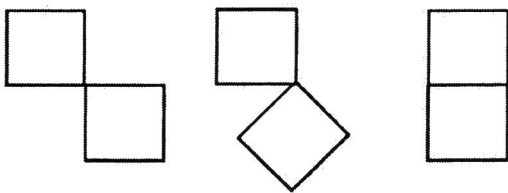
7. What number should place the question mark ?



- (1) 64 (2) 63 (3) 72 (4) 78
8. In the question, some letters are missing shown by (____). The missing letters are given in proper sequence as one of the five alternatives given under each question. Find out the correct alternative.
C ____ BCCD ____ CCDB ____ CDBCC ____ BC
(1) DBCD (2) DBDD (3) BDAA (4) BDCD
9. In the following question, three statements of numbers following same rules are given. Find the rule and accordingly find the value of the number ?
If $84 \oplus 72 = 45$; $63 \oplus 41 = 33$; $25 \oplus 52 = 33$, then $94 \oplus 82 = ?$
(1) 45 (2) 59 (3) 56 (4) 65
10. Choose the correct mirror image of fig. (X).

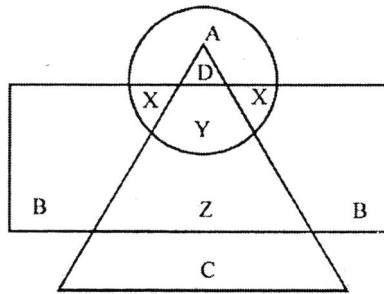


11. What comes next in the sequence ?



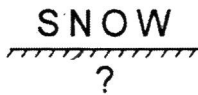
12. Manish is fourteenth from the right end in a row of 40 boys. What is his position from the left end?
(1) 24th (2) 25th (3) 26th (4) 27th
13. Give the next number in this series :
220, 200, 100, 80, 40, 20, ?
(1) 20 (2) 10 (3) 30 (4) 40

14. In the following diagram, the circle represents college Professors, the triangle stands for surgical Specialists, and medical Specialists are represented by the rectangle



College professors who are also surgical Specialists are represented by

- (1) A (2) B (3) C (4) D
15. Find the correct option for the water -image for the following



- (1) 2NOM (2) SNOM (3) SNOM (4) SNOW
16. Pointing to Kapil, Shilpa said, "His mother's brother is the father of my son Ashish." How is Kapil related to Shilpa ?
- (1) Sister-in-law (2) Nephew (3) Mother (4) Mother-in-law
17. If CAT is 48, Z is 52, then what is TEA equal to :
- (1) 48 (2) 52 (3) 60 (4) 50
18. A cube painted red on two adjacent faces and black on the faces opposite to the red faces and green on the remaining faces is cut into sixty-four smaller cubes of equal size.
- How many cubes are there which have no face painted ?
- (1) 0 (2) 4 (3) 8 (4) 16
19. A person starts from a point A and travels 3 km eastwards to B and then turns left and travels thrice that distance to reach C. He again turns left and travels five times the distance he covered between A and B and reaches his destination D. The shortest distance between the starting point and the destination is :
- (1) 12 km (2) 15 km (3) 16 km (4) 18 km



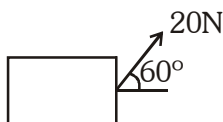
If the mirror image of the figure is rotated to 90° in clockwise direction, it will look like :



PART-II
SECTION-A : PHYSICS

This section contains **15 Multiple Choice Questions**. Each question has four choices (1), (2), (3) and (4) out of which ONLY ONE is correct.

- 21.** The unit of force is
 (1) Newton (2) $\text{kg m}^2 / \text{s}^2$ (3) $\text{kg m} / \text{s}^2$ (4) both 1 and 3
- 22.** A particle is moving with a velocity of 2 m/s subjected to have a uniform acceleration of 2.5 m/s^2 . The distance covered in 4 seconds will be ?
 (1) 8 m (2) 28 m (3) 12 m (4) 5 m
- 23.** When a person move from equator to pole, the weight of person
 (1) increases (2) decreases
 (3) remain same (4) first increase then decrease
- 24.** As in the figure, a force of 20 N is applied on an object produces displacement of 2 m. The work done by force is



- (1) 40 J (2) 20 J (3) 10 J (4) $20\sqrt{3} \text{ J}$
- 25.** A canon after firing recoils due to
 (1) conservation of energy (2) Newton's first law of motion
 (3) Newton's third law of motion (4) Newton's second law of motion
- 26.** An object is moving with uniform velocity must
 (1) have a net force unit (2) have force of gravity zero
 (3) stop due to gravity (4) have net force zero on it
- 27.** A person start moving in east after travelling 10 m take a left turn and move 6 m then again take a left turn and move 2m the displacement of person is :
 (1) 8 m (2) 18 m (3) 10 m (4) 100 m
- 28.** The unit of relative density is
 (1) kg / m^3 (2) g / cm^3 (3) both (4) unitless
- 29.** What kind of energy is present in a rotating wheel ?
 (1) kinetic energy (2) electrical energy (3) potential energy (4) wind energy
- 30.** If equal force act in a body A and B, and mass of body A is 5 times that of mass of B. The acceleration of body B with A will be
 (1) five time that of A (2) 1 / 5 time that of A
 (3) both body have same acceleration (4) 25 times that of A
- 31.** There are two masses ' m_1 ' and ' m_2 ' separated by a distance 'r'. If one of the mass becomes doubled and separation between massess half, the new force between masses will be
 (1) The force remain same (2) become 4 times
 (3) become 8 times (4) become 16 times
- 32.** If kinetic energy of an object is increased by 100%, then percentage change in momentum of object will be :
 (1) 100% (2) 50% (3) 200% (4) 41.4%

- 33.** The energy consumed in Kwh in 6 hours by two devices of power 600 W each :
 (1) 72000 (2) 3600 (3) 7.2 (4) 7200
- 34.** Two racing cars of masses m_a and m_b are moving on circles r_a and r_b respectively. Their speed are such that they will complete circle in equal length of time 'T'. The ratio of angular velocity of car B to car A is ?
 (1) 1 : 1 (2) $r_b : r_a$ (3) $r_a : r_b$ (4) $\frac{r_b^2}{r_a^2}$
- 35.** If R is the radius of Earth, the height at which the weight of body becomes $1/4^{\text{th}}$ of its weight on the surface of the Earth is :
 (1) R (2) $\frac{R}{2}$ (3) 2R (4) $\frac{R}{4}$

SECTION-B : CHEMISTRY

This section contains **15 Multiple Choice Questions**. Each question has four choices (1), (2), (3) and (4) out of which ONLY ONE is correct.

- 36.** The correct order which describes the true solution colloidal solution, true solution, and suspension in the order of their increasing stability is
 (1) Suspension < colloidal solution < true solution
 (2) colloidal solution < true solution < suspension
 (3) true solution < colloidal solution < suspension
 (4) colloidal solution < suspension < true solution
- 37.** 2 moles of H_2 burn to form 2 moles of H_2O . The mass of Oxygen used up will be
 (1) 16 g (2) 32g (3) 2 moles (4) 4 moles
- 38.** Isobars do not differ in the number of ____.
 (1) protons (2) electrons (3) neutrons (4) nucleons
- 39.** A few pieces of zinc were treated with dilute sulphuric acid in a test tube. Which of the following observations is correct ?
 (1) It is a physical change and zinc dissolves
 (2) It is a chemical change and zinc dissolves
 (3) It is a chemical change, zinc dissolves and H_2 gas is evolved
 (4) It is a physical change and H_2 gas is evolved
- 40.** Law of constant proportions was put forward by
 (1) Dalton (2) Proust (3) Lavoisier (4) Beryllius
- 41.** Which of the following is the correct symbol for copper ?
 (1) Co (2) Cr (3) Cu (4) C
- 42.** Out of the given four mixtures, the one that appears clear and transparent in
 (1) Chalk powder and water (2) Sand and water
 (3) Starch and water (4) Common salt and water
- 43.** The chemical symbol for nitrogen gas is
 (1) Ni (2) N_2 (3) N^+ (4) N
- 44.** Which of the following contains maximum number of molecules ?
 (1) 1g of CO_2 (2) 1 g N_2 (3) 1g of H_2 (4) 1g CH_4

- 45.** The valency of which of the following elements is zero ?
 (1) Hydrogen (2) Helium (3) Gold (4) Silver
- 46.** Scattering of light occurs when a beam of light is passed through
 (1) Blood (2) Water
 (3) Copper sulphate solution (4) Brine
- 47.** Mass of one atom of Oxygen is
 (1) $\frac{16}{6.023 \times 10^{23}}$ g (2) $\frac{32}{6.023 \times 10^{23}}$ g (3) $\frac{1}{6.023 \times 10^{23}}$ g (4) 8 u
- 48.** The ion of an element has three positive charges. Mass number of the atom is 27 and the number of neutrons is 14. What is the number of electrons in the ion ?
 (1) 13 (2) 10 (3) 14 (4) 16
- 49.** Which of the following mixtures cannot be separated by fractional distillation ?
 (1) Acetone and methyl alcohol (2) Chloroform and Benzene
 (3) Water(5%) and ethyl alcohol (95%) (4) Benzene and Toulene
- 50.** The atomic number of an element is 15. Hence it will show a valency of
 (1) 3 only (2) 5 only (3) both 3 and 5 (4) neither 3 nor 5

SECTION-C : BIOLOGY

This section contains **15 Multiple Choice Questions**. Each question has four choices (1), (2), (3) and (4) out of which ONLY ONE is correct.

- 51.** The jelly-like substance excluding the nucleus in the cell is called-
 (1) Cytoplasm (2) Protoplasm (3) Nucleoplasm (4) Cell plasm
- 52.** The tissue which help in increasing the girth (diameter) of plant is-
 (1) Apical meristem (2) Lateral meristem
 (3) Intercalary meristem (4) None of the above
- 53.** The method of immunisation was first tried by _____
 (1) Edward Jenner (2) Robert Hooke (3) Louis Hook (4) Louis Pasteur
- 54.** The term tissue was given by-
 (1) Meyer (2) Leeuwenhoek (3) Robert Hooke (4) Bichat
- 55.** Nucleus is an important component of the living cell.
 Find correct statements regarding nucleus-
 (i) It is generally spherical and located in the centre of the cell.
 (ii) It can be stained and seen easily with the help of a microscope.
 (iii) Nucleus is separated from the protoplasm by a membrane called the nuclear membrane.
 (iv) Nuclear membrane is non porous and allows the movement of materials between the cytoplasm and the inside of the nucleus.
 (v) Smaller spherical body in the nucleus is called the nucleolus.
 (vi) In addition, nucleus contains thread-like structures called ribosomes.
 (vii) Genes help in inheritance of characters from the parents to the offspring.
 (viii) The chromosomes in the nucleus can be seen only when the cell grow.
 (ix) Nucleus also acts as control centre of the activities of the cell.
 (1) (i), (iii), (v), (vii) and (ix) (2) (ii), (iv), (vi) and (viii)
 (3) (iii), (v), (vi), (viii) and (ix) (4) (i),(ii),(v), (vii) and (ix)

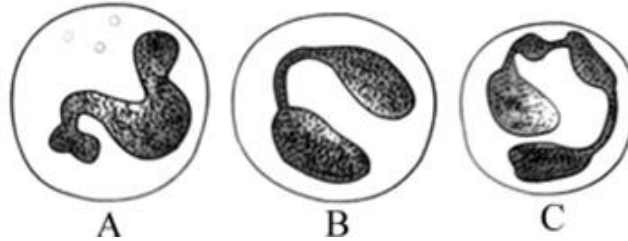
56. The bacteria among the following is

- (1) Plasmodium (2) Trypanosoma (3) Rabies virus (4) Salmonella typhi

57. Protein factory of cell are-

- (1) Mitochondria (2) Ribosomes (3) Lysosomes (4) Chloroplast

58. Identify the leucocytes given below-

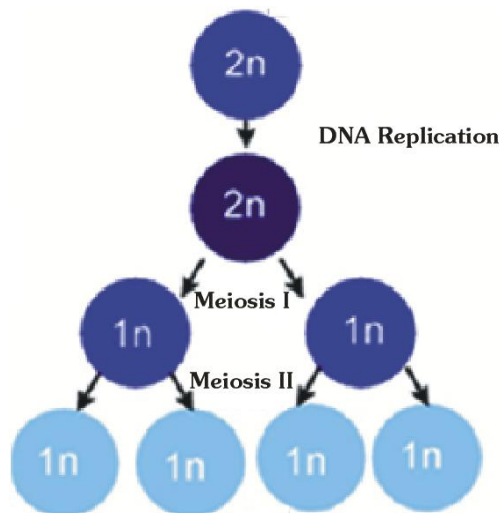


- (1) A- eosinophil, B-monocyte, C-basophil (2) A -basophil , B - lymphocyte, C-neutrophil,
(3) A - neutrophil, B-basophil , C- eosinophil (4) A -basophil , B -eosinophil, C- neutrophil

59. Sleeping sickness is caused by (i) _____ , which is a (ii) _____.

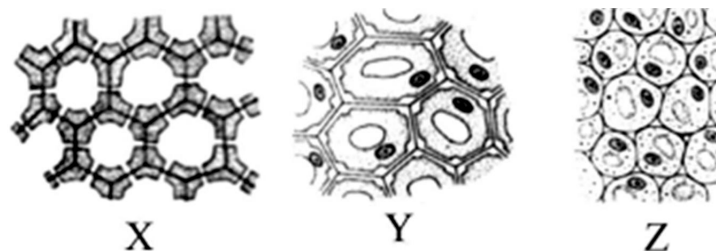
- (1) (i) Leishmania, (ii) protozoan (2) (i) Trypanosoma, (ii) protozoan
(3) (i) Staphylococcus, (ii) bacterium (4) (i) Trypanosoma, (ii) bacterium

60. Following figure shows a type of cell division. Which statement is not correct about it-



- (1) It helps in formation of spores (2) It occurs in somatic cells
(3) It occurs in gamete forming cells (4) It brings about variations

61. Choose the correct answer for following figure-



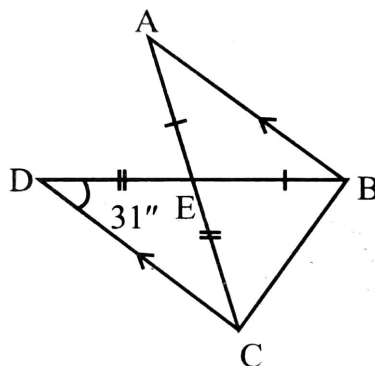
- (1) X-living mechanical tissue, Y-dead mechanical tissue, Z-storage tissue
(2) X-dead mechanical tissue, Y-living mechanical tissue, Z-storage tissue
(3) X-living mechanical tissue, Y-storage tissue, Z- dead mechanical tissue
(4) X-storage tissue , Y-dead mechanical tissue, Z -living mechanical tissue

- 62.** The only connective tissue without fibroblasts is-
 (1) areolar connective tissue (2) bone
 (3) cartilage (4) blood
- 63.** Congenital disease is
 (1) Deficiency disease (2) Present from the time of birth
 (3) Spread from man to man (4) That occurs during lifetime
- 64.** Sporozoites of the malarial parasites are found in
 (1) the saliva of female Anopheles mosquito, which is freshly moulted
 (2) the saliva of infected female Anopheles mosquito
 (3) RBCs of an infected human
 (4) the spleen of an infected human
- 65.** What part of the cell serves as the intracellular highway?
 (1) Cell membrane (2) Endoplasmic reticulum
 (3) Golgi apparatus (4) Mitochondria

SECTION-D : MATHEMATICS

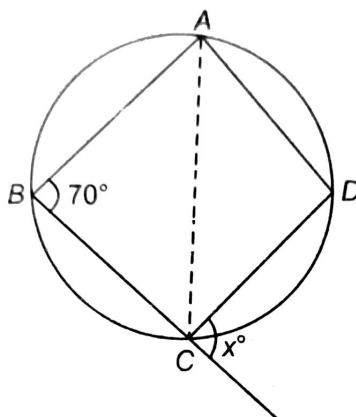
This section contains **15 Multiple Choice Questions**. Each question has four choices (1), (2), (3) and (4) out of which ONLY ONE is correct.

- 66.** Two cubes of sides 6 cm each are kept side by side to form a rectangular parallelopoid. The area in (sq. cm) of the whole surface of the rectangular parallelopoid is
 (1) 432 (2) 360 (3) 396 (4) 340
- 67.** In the given figure, $AB \parallel DC$, $\triangle EDC$ and $\triangle EBA$ are both isosceles triangles and $\angle EDC = 31^\circ$.



Identify the measure of $\angle AED$.

- (1) 118° (2) 62° (3) 57° (4) 89°
- 68.** In cyclic quadrilateral ABCD, $AD = DC$, $AB = BC$ and $\angle ABC = 70^\circ$. Value of x is

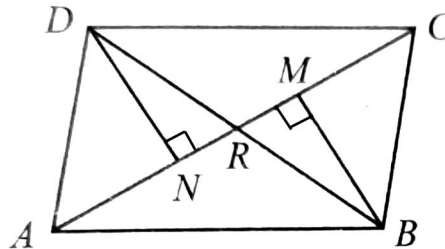


- (1) 35° (2) 55° (3) 90° (4) 110°

69. If $k = 2 - x$, then the value $x^3 + 6kx + k^3 - 8$ is

- (1) 1 (2) -3 (3) 0 (4) -2

70. In quadrilateral ABCD, BM and DN are drawn perpendiculars to AC such that $BM = DN$. If $BR = 8$ cm, then BD is



- (1) 4 cm (2) 2 cm (3) 12 cm (4) 16 cm

71. The mean of 50 numbers is 40. It was found that three numbers 23, 29 and 20 were wrongly taken as 28, 92 and 2. Find the correct mean.

- (1) 41 (2) 38 (3) 42 (4) 39

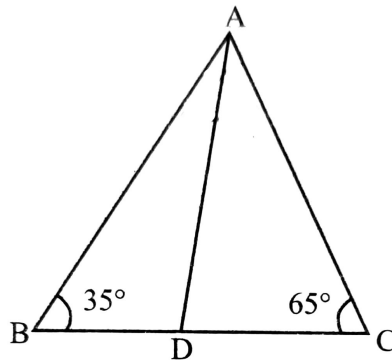
72. Three sides of triangle are $a = 10$ cm, $b = 20$ cm and $c = 20$ cm. If a and b are increased by 50%, then the percentage change in semi-perimeter of triangle is

- (1) 20% increase (2) 30% increase (3) 75% increase (4) No change

73. The distance between the graph of the equations $y = 1$ and $y = 4$ is

- (1) 2 (2) 5 (3) 1 (4) 3

74. In $\triangle ABC$, $\angle B = 35^\circ$, $\angle C = 65^\circ$ and the bisector AD of $\angle BAC$ meets BC at D. Then, which of the following is true ?



- (1) $AD > BD > CD$ (2) $BD > AD > CD$ (3) $AD > CD > BD$ (4) None of these

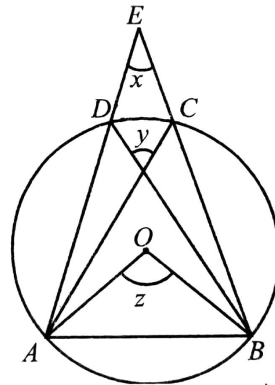
75. Three coins are tossed together. Then the probability of getting at least one head is

- (1) $\frac{1}{2}$ (2) $\frac{3}{4}$ (3) $\frac{1}{8}$ (4) $\frac{7}{8}$

76. $\sqrt{m^4 n^4} \times \sqrt[6]{m^2 n^2} \times \sqrt[3]{m^2 n^2} = (mn)^k$, then find the value of k .

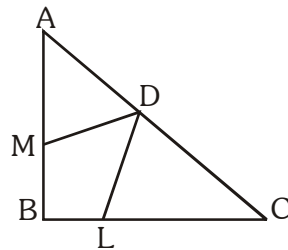
- (1) 6 (2) 3 (3) 2 (4) 1

77. In figure, O is the centre of the circle, then :



- (1) $x + y + z = 180^\circ$ (2) $z = 2x + y$ (3) $y = z + x$ (4) $z = x + y$

78. In the adjoining figure, $\triangle ABC$ is right angled at B and $CD = CL$ and $AM = AD$. Calculate $\angle MDL$.



- (1) 30° (2) 15° (3) 45° (4) 60°

79. Positive integers from 1 to 21 arranged in 3 groups of 7 integers each, in some particular order. Then the highest possible mean of the medians of these 3 groups is

- (1) 16 (2) 12.5 (3) 11 (4) 14

80. The area of triangle formed by the lines $x = 0$, $y = 0$ and $\frac{x}{a} + \frac{y}{b} = 1$, is :

- (1) ab (2) $\frac{ab}{2}$ (3) $2ab$ (4) $\frac{ab}{3}$

SPACE FOR ROUGH WORK